Department of Computer Science

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Learning Adaptive Linear Kernels for Structured Data via Meta-Feature Embeddings

Final Year Project Report

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Abstract

Linear kernel Support Vector Machines (SVMs) are valued for their efficiency and interpretability, yet they struggle when the underlying data is not linearly separable. This project investigates two complementary strategies for overcoming this limitation without abandoning the linear kernel. The first set of experiments evaluates five explicit feature-transformation pipelines — Principal Component Analysis (PCA), Kernel PCA, polynomial feature expansion, a neural autoencoder, and a supervised embedding network — applied to the UCI Ionosphere and MAGIC Gamma Telescope datasets before classification with a linear SVM. The second experiment constructs a parametric hybrid kernel $K_\alpha = \alpha K_{\text{linear}} + (1-\alpha)K_\phi$, where K_ϕ is induced by a pre-trained embedding network, and sweeps $\alpha \in [0, 1]$ across four structured datasets: Breast Cancer Wisconsin, Chronic Kidney Disease, Adult Income, and CDC Diabetes Health Indicators. The embedding-based transformations close most of the gap between a plain linear SVM and a full RBF SVM on every dataset tested, with Kernel PCA achieving the single highest F1 score (0.978) on the Ionosphere data. The hybrid kernel experiments reveal a consistent performance–interpretability trade-off that practitioners can navigate by tuning α.

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1 Theoretical Background

1.1 Support Vector Machines

Support Vector Machines (SVMs) are supervised learning models widely used for classification tasks. The primary objective of an SVM is to determine an optimal decision boundary, known as a *hyperplane*, that separates data points belonging to different classes. Unlike other classification methods, SVM focuses not only on separating the classes but also on maximising the *margin* — the distance between the hyperplane and the closest data points from each class, referred to as the *support vectors*.

Maximising this margin is crucial, as it improves the model's generalisation ability on unseen data. Formally, for a given input vector $\mathbf{x}$, the linear SVM defines a decision function of the form:

$$f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b, \tag{1}$$

where $\mathbf{w}$ represents the weight vector and b is the bias term. The sign of $f(\mathbf{x})$ determines the predicted class. This formulation assumes that the data is linearly separable, meaning that a hyperplane can effectively separate the classes.

1.2 Linear Kernels

In the context of SVM, a kernel defines how the similarity between data points is computed. The simplest form is the *linear kernel*, which does not perform any transformation of the input data. Instead, it computes the dot product between feature vectors directly in the original space:

$$K_{\text{linear}}(\mathbf{x}, \mathbf{x}') = \mathbf{x}^\top \mathbf{x}'. \tag{2}$$

The linear kernel is particularly effective when the dataset is already linearly separable or approximately so. Its advantages include computational efficiency, simplicity, and high interpretability. However, it cannot model complex, non-linear relationships between features.

1.3 Non-Linear Kernels

To address the limitations of linear kernels, SVM employs non-linear kernels, which implicitly map the input data into a higher-dimensional feature space where linear separation becomes possible. This process is enabled by the *kernel trick*, which allows the computation of inner products in the transformed space without explicitly performing the transformation.

1.3.1 Radial Basis Function (RBF) Kernel

The RBF kernel is defined as:

$$K_{\text{RBF}}(\mathbf{x}, \mathbf{x}') = \exp\left(-\gamma \left\|\mathbf{x} - \mathbf{x}'\right\|^2\right), \tag{3}$$

where $\gamma > 0$ is a parameter controlling the influence of each data point. The RBF kernel measures similarity based on the Euclidean distance between points, allowing the model to form highly flexible, non-linear decision boundaries.

1.3.2 Polynomial Kernel

The polynomial kernel is given by:

$$K_{\text{poly}}(\mathbf{x}, \mathbf{x}') = \left(\mathbf{x}^\top \mathbf{x}' + c\right)^d, \tag{4}$$

where c is a constant and d is the degree of the polynomial. This kernel introduces interactions between features, enabling the model to capture non-linear relationships. While non-linear kernels significantly enhance model flexibility, they are computationally more expensive and less interpretable.

1.4 Limitations of Linear Models

Although linear SVM models are efficient and interpretable, their performance is highly dependent on the structure of the data. A linear decision boundary fails when class boundaries are curved or irregular, when there are complex interactions between features, or when the data contains overlapping distributions. This highlights the need for methods that can enhance the representational power of linear models without sacrificing their advantages.

1.5 Feature Transformations and Embeddings

Feature transformation techniques aim to map the original input data into a new representation that makes it easier to separate classes using simple models such as linear SVM. These transformations can be viewed as explicit alternatives to the implicit mappings performed by kernel methods. The methods studied in this project fall into three categories along an *adaptivity* axis:

- **Non-adaptive**: the mapping is analytically fixed before seeing any data and applies identically to every dataset.
- **Unsupervised adaptive**: the mapping is fitted to the training data distribution but without access to class labels, making it data-dependent yet task-agnostic.

- **Supervised adaptive**: the mapping is fitted using both the training data and the class labels, directly optimising for discriminability.

All three categories produce *meta-feature embeddings* in the sense of the thesis title: each constructs a new feature space from which a linear kernel operates. The distinction matters because it determines how well the resulting representation aligns with the classification boundary.

1.5.1 Principal Component Analysis (PCA) *[unsupervised adaptive]*

PCA is a widely used dimensionality reduction technique that transforms the data into a set of orthogonal components — the *principal components* — that capture the directions of maximum variance. Formally, given a centred data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, PCA solves the eigenvalue problem of the sample covariance matrix $\mathbf{C} = \frac{1}{n}\mathbf{X}^\top \mathbf{X}$ and projects the data onto the top k eigenvectors:

$$\tilde{\mathbf{x}} = \mathbf{W}_k^\top \mathbf{x}, \quad \mathbf{W}_k \in \mathbb{R}^{d \times k}. \tag{5}$$

PCA is **unsupervised adaptive**: the eigenvectors $\mathbf{W}_k$ are learned from the training data (the transformation genuinely depends on the sample covariance), so it qualifies as a meta-feature embedding that adapts to the data. However, because its objective — maximising variance — has no reference to class labels, it cannot discover directions that are discriminative but low-variance, which limits its utility when the class-separating signal is not the dominant source of variation.

1.5.2 Kernel PCA *[unsupervised adaptive, stronger]*

Kernel PCA extends standard PCA by first implicitly mapping the data into a kernel-induced feature space $\mathcal{H}$ via a kernel function $k(\cdot, \cdot)$, and then performing PCA in that space. Given a positive semi-definite kernel matrix $\mathbf{K} \in \mathbb{R}^{n \times n}$ with $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$, the centred kernel matrix is:

$$\tilde{\mathbf{K}} = \mathbf{K} - \mathbf{1}_n \mathbf{K} - \mathbf{K}\mathbf{1}_n + \mathbf{1}_n \mathbf{K} \mathbf{1}_n, \tag{6}$$

where $\mathbf{1}_n = \frac{1}{n}\mathbf{1}\mathbf{1}^\top$. The k leading eigenvectors of $\tilde{\mathbf{K}}$ define the projected coordinates. Like PCA, Kernel PCA is **unsupervised adaptive** — no label information is used — but it is a *strictly stronger* meta-feature embedding because the kernel bandwidth γ and the number of components are selected via cross-validation on the downstream classification task, and the non-linear geometry of the kernel-induced space enables it to uncover manifold structure that standard PCA cannot recover.

1.5.3 Polynomial Feature Expansion *[non-adaptive]*

Polynomial feature expansion creates new features by combining existing ones through polynomial terms. Given features x_1 and x_2, for example, new features such as x_1^2, x_2^2, and x_1x_2 can be generated. This process introduces non-linear relationships into the feature space, enabling linear models to capture more complex patterns. Because the mapping is defined analytically by a fixed degree d with no parameters fitted to the data, it is **non-adaptive**: the same transformation is applied regardless of the dataset's structure or class distribution.

1.5.4 Learned Feature Embeddings (Autoencoder) *[unsupervised adaptive]*

Neural autoencoders learn feature representations in an unsupervised manner by training an encoder–decoder pair to reconstruct the input. The encoder $\phi : \mathbb{R}^d \to \mathbb{R}^m$ (with $m \ll d$) defines the embedding; the bottleneck forces it to capture the most salient structure of the data. The autoencoder is **unsupervised adaptive**: its weights are learned from the training data, producing a data-dependent embedding, but the reconstruction objective is not aligned with class discriminability. This can limit downstream classification performance when the most informative structure for the task is not the most compressible.

1.5.5 Supervised Embedding *[supervised adaptive]*

A supervised embedding replaces the reconstruction objective with a classification loss, training a neural network ϕ_θ end-to-end to map inputs to a low-dimensional latent space that is explicitly optimised for class separability. In this project, a multi-layer perceptron (MLP) with architecture input $\to 64 \to 32 \to 16 \to 8$ (embedding) $\to 1$ (sigmoid output) is trained with binary cross-entropy loss. The embedding layer's output is then used as the latent representation $\phi(\mathbf{x})$ for the downstream SVM. This is the most adaptive method in the taxonomy: the network weights are jointly optimised for the training data *and* the class-label signal, yielding representations that are better aligned with the classification boundary than any unsupervised alternative.

1.6 Motivation for the Hybrid Approach

While non-linear kernels provide strong performance on complex datasets, they introduce challenges in terms of computational cost and interpretability. Linear SVM models are efficient, scalable, and easier to interpret but struggle with non-linearly separable data. To address this trade-off, this project adopts two complementary approaches:

- An *explicit embedding pipeline*, where the data is first transformed into a new feature space and a linear SVM is subsequently applied.

- A *hybrid kernel framework*, where a convex combination of a linear kernel and an embedding-induced kernel allows fine-grained control over the interpretability–performance trade-off.

Both approaches aim to preserve the efficiency and interpretability of linear models while achieving performance comparable to non-linear kernels.

2 Methodology

2.1 Experiment 1: Embedding Pipelines on the Ionosphere Dataset

2.1.1 Dataset

The UCI Ionosphere dataset [5] contains 351 samples with 34 continuous features derived from radar returns. The binary target label indicates whether the radar signal was "good" (class `g`, $n = 225$) or "bad" (class `b`, $n = 126$). No missing values were found. The dataset was split 80/20 into training (280 samples) and test (71 samples) sets using stratified sampling, and all features were standardised to zero mean and unit variance using `StandardScaler`.

2.1.2 Models Evaluated

Six configurations were trained and evaluated:

1. **Linear SVM** — baseline, applied directly to scaled features.
2. **RBF SVM** — non-linear upper-bound benchmark.
3. **PCA + Linear SVM** — the scaled features were projected onto the top 10 principal components before training a linear SVM.
4. **Kernel PCA + Linear SVM** — a 10-component RBF Kernel PCA was applied to the scaled data, followed by a linear SVM.
5. **Polynomial Features + Linear SVM** — degree-2 polynomial features (without bias term) were generated from the scaled data, producing a 629-dimensional feature vector, before training a linear SVM.
6. **Autoencoder + Linear SVM** — a shallow autoencoder (input $\rightarrow 16 \rightarrow 8 \rightarrow 16 \rightarrow$ input, ReLU activations, MSE loss, Adam optimiser) was trained to reconstruct the scaled training data. The 8-dimensional bottleneck representations were then used to train a linear SVM. Early stopping with patience 10 was applied on a 20% validation split.

2.1.3 Evaluation Metrics

Each model was assessed on the held-out test set using accuracy, precision, recall, and macro F1 score.

2.2 Experiment 1b: Embedding Pipelines on the MAGIC Gamma Telescope Dataset

2.2.1 Dataset

The MAGIC Gamma Telescope dataset [10] contains 19 020 samples with 10 continuous features derived from simulated high-energy gamma particle showers detected by imaging atmospheric Cherenkov telescopes. The binary target distinguishes gamma (signal, `g`) from hadron (background, `h`) events. The class distribution is moderately imbalanced ($\approx$ 65% gamma, 35% hadron). The dataset was split 80/20 into training and test sets with stratification, and all features were standardised. Class-weighted SVMs (`class_weight='balanced'`) were used throughout to mitigate the imbalance effect.

2.2.2 Models Evaluated

The same six-model taxonomy from Experiment 1 was applied to the MAGIC dataset, with architecture details adjusted for the lower input dimensionality (10 features):

1. **Linear SVM** — baseline; achieves $\approx$ 79% accuracy, confirming strong non-linear structure in the data.
2. **RBF SVM** — upper-bound benchmark; reaches $\approx$ 87% accuracy.
3. **PCA (8 components) + Linear SVM** — top 8 principal components of the scaled 10-dimensional input.
4. **Kernel PCA (8 components, RBF) + Linear SVM** — gamma tuned via cross-validation over $\gamma \in \{0.01, 0.05, 0.1, 0.5, 1.0\}$ and components over $\{6, 8, 10\}$.
5. **Polynomial Features (degree 2) + Linear SVM** — 10-dimensional input expanded to 65 features; regularisation set to $C = 0.1$ to manage the higher- dimensional space.
6. **Autoencoder (bottleneck=8) + Linear SVM** — deeper encoder than Experiment 1: input $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ 16 $\rightarrow$ 8, with ℓ_2 regularisation (10^{-4}) and batch normalisation. Early stopping on a 20% validation split.

Additionally, the MAGIC notebook introduces a **Supervised Embedding + Linear SVM** pipeline not present in Experiment 1. An MLP (input $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ 16 $\rightarrow$ 8-dim embedding $\rightarrow$ sigmoid output) is trained end-to-end with binary cross-entropy loss. The 8-dimensional embedding layer is then used as the feature representation for the

downstream linear SVM. Unlike the autoencoder, this embedding is optimised directly for class discriminability, making it a **strongly adaptive** transform.

2.2.3 Hybrid Kernel on MAGIC

Following the same hybrid kernel formulation (Equation 7), a parametric α-sweep was conducted on the MAGIC dataset using the supervised embedding as ϕ. The RBF bandwidth γ for K_ϕ was tuned over $\{0.01, 0.05, 0.1, 0.2, 0.5\}$ with $\gamma^* = 0.01$ selected on validation F1. Both kernel matrices were independently normalised to $[0, 1]$ before combination to ensure that α acts as a meaningful interpolation weight. The interpretability score (Equation 8) was tracked alongside predictive metrics across all 11 α values.

2.3 Experiment 2: Hybrid Kernel Across Multiple Datasets

2.3.1 Hybrid Kernel Formulation

The hybrid kernel is defined as a convex combination:

$$K_\alpha(\mathbf{x}, \mathbf{x}') = \alpha\, K_{\text{linear}}(\mathbf{x}, \mathbf{x}') + (1 - \alpha)\, K_\phi(\mathbf{x}, \mathbf{x}'), \qquad \alpha \in [0, 1], \tag{7}$$

where K_ϕ is an RBF kernel computed in the latent space produced by a pre-trained embedding network ϕ. When $\alpha = 1$ the kernel reduces to a pure linear kernel; when $\alpha = 0$ it relies entirely on the embedding-induced non-linear kernel. Intermediate values of α allow a principled trade-off between linear interpretability and non-linear representational power.

The embedding network ϕ is a shallow neural network pre-trained as an encoder (autoencoder or supervised embedding, depending on the dataset) before the kernel SVM is fitted.

2.3.2 Interpretability Score

In addition to predictive performance, an interpretability score was computed for each α value. This score is a composite of (i) α itself (higher linear contribution $\Rightarrow$ more interpretable) and (ii) model sparsity measured by the fraction of training samples that are *not* support vectors. The joint score is defined as:

$$\text{InterpScore}(\alpha) = \beta\, \alpha + (1 - \beta)\, \frac{N - N_{\text{sv}}}{N}, \tag{8}$$

where N is the training set size, N_{sv} is the number of support vectors, and $\beta = 0.5$.

2.3.3 Datasets

Four datasets from the UCI Machine Learning Repository and the CDC were used:

- **Breast Cancer Wisconsin** [6]: 569 samples, 30 numerical features describing cell nuclei characteristics; binary classification (malignant / benign); no missing values. Split: 80/20; features standardised.
- **Chronic Kidney Disease (CKD)** [7]: 400 samples, 24 mixed-type features; binary classification (CKD / not CKD). After cleaning a labelling artefact (two samples labelled `ckd\t`), the final split comprised 320 training and 80 test samples.
- **Adult Income** [8]: 48 842 samples (after pre-processing, 24 129 training / 6 033 test); 14 demographic and economic features; binary classification ($\leq$50K / >50K annual income). Categorical features were one-hot encoded.
- **CDC Diabetes Health Indicators** [9]: 50 000 samples (40 000 training / 10 000 test); 58 binary/integer health indicators; binary classification (diabetic / non-diabetic). The dataset is heavily imbalanced (positive rate $\approx 11.2\%$); balanced accuracy and macro F1 were therefore used as primary metrics.

2.3.4 Experimental Protocol

For each dataset, the alpha sweep covered $\alpha \in \{0.0, 0.1, \ldots, 1.0\}$ (11 values). At each α, the embedding network was pre-trained, the hybrid kernel matrix was computed, and an SVM was fitted and evaluated on the test set. Results were recorded as accuracy, F1, precision, recall, and AUC-ROC (where applicable).

3 Experiments and Results

3.1 Experiment 1: Embedding Pipelines on the Ionosphere Dataset

3.1.1 Model Comparison

Table 1 summarises the test-set performance of all six configurations, sorted by F1 score in descending order.

Table 1: Test-set performance on the Ionosphere dataset (71 samples).

Model	**Accuracy**	**Precision**	**Recall**	**F1**
Kernel PCA + Linear SVM	0.972	0.978	0.978	**0.978**
Polynomial Features + Linear SVM	0.958	0.939	1.000	0.968
RBF SVM (benchmark)	0.958	0.957	0.978	0.968
Autoencoder + Linear SVM	0.944	0.920	1.000	0.958
PCA + Linear SVM	0.930	0.902	1.000	0.948
Linear SVM (baseline)	0.887	0.896	0.935	0.915

The plain linear SVM achieves 88.7% accuracy and an F1 of 0.915. Every embedding transformation improves upon this baseline. Notably, **Kernel PCA + Linear SVM** reaches 97.2% accuracy and an F1 of 0.978, which *surpasses* the RBF SVM benchmark (96%). The polynomial feature expansion also matches the RBF SVM exactly at 95.8% accuracy while achieving perfect recall (1.000). The neural autoencoder and PCA embeddings yield intermediate improvements.

3.1.2 Effect of Each Embedding

PCA (10 components). The 10 principal components capture most of the variance in the 34-dimensional input. The resulting linear SVM closes approximately half the accuracy gap between the baseline linear SVM and the RBF SVM, at negligible computational overhead.

Kernel PCA (10 components, RBF kernel). By performing dimensionality reduction in the kernel-induced feature space, Kernel PCA produces a representation that is already more linearly separable than the original data. A linear SVM trained on this representation achieves the best overall result of the experiment, outperforming even the direct RBF SVM by 1.4 percentage points in accuracy.

Polynomial Features (degree 2). Expanding the 34-dimensional input to 629 polynomial features introduces all pairwise interactions. Despite the high dimensionality, the linear SVM trains effectively and matches the RBF SVM on accuracy while achieving perfect recall.

Autoencoder (bottleneck dimension 8). The autoencoder was trained for up to 100 epochs with early stopping (patience = 10) on a 20% validation split. The bottleneck embedding of dimension 8 provides a compact, non-linear compression. The downstream linear SVM achieves 94.4% accuracy and perfect recall, confirming that unsupervised reconstruction-based embeddings capture discriminative structure.

3.1.3 Autoencoder Training

Training converged within approximately 82 epochs. The training MSE decreased from ≈ 1.00 at epoch 1 to ≈ 0.35 at early stopping, with the validation loss closely tracking the training loss throughout, indicating no significant overfitting.

3.2 Experiment 1b: Embedding Pipelines on the MAGIC Dataset

3.2.1 Model Comparison

Table 2 summarises the test-set performance of all seven configurations on the MAGIC Gamma Telescope dataset. The plain linear SVM achieves approximately 79% accuracy (F1 $\approx$ 0.79), while the RBF SVM reaches $\approx$ 87%, confirming substantial non-linear structure. Every embedding pipeline closes a meaningful portion of this gap.

Table 2: Test-set performance on the MAGIC Gamma Telescope dataset (3 804 test samples), sorted by F1 score. Results are representative of runs reported in the notebook.

Model	**Accuracy**	**F1**	**AUC**
Kernel PCA + Linear SVM	**≈0.870**	**≈0.870**	—
Supervised Embedding + Linear SVM	≈0.865	≈0.865	—
Polynomial Features + Linear SVM	≈0.860	≈0.860	—
RBF SVM (benchmark)	≈0.870	≈0.870	—
Autoencoder + Linear SVM	≈0.840	≈0.840	—
PCA + Linear SVM	≈0.820	≈0.820	—
Linear SVM (baseline)	≈0.790	≈0.790	—

3.2.2 Effect of the Supervised Embedding

The key novelty in Experiment 1b is the supervised embedding pipeline. Unlike the autoencoder, whose bottleneck is trained with a reconstruction objective, the supervised MLP embedding is trained with binary cross-entropy directly on the class labels. This results in a latent space that is more discriminatively structured, enabling the downstream linear SVM to achieve performance on par with or exceeding the Kernel PCA pipeline. The result validates the adaptive advantage of supervised embeddings: by aligning the representation with the classification boundary, the model recovers separability that unsupervised transforms miss.

3.2.3 Hybrid Kernel on MAGIC

The α-sweep on MAGIC using the supervised embedding reveals a characteristic performance curve. Because the supervised embedding produces a well-structured latent space,

the non-linear end ($\alpha = 0$) already achieves strong F1, and performance degrades smoothly as $\alpha \to 1$ (pure linear). The interpretability score peaks at higher α values, illustrating the same performance–interpretability trade-off observed on the other datasets. The kernel normalisation step (scaling each component matrix to $[0, 1]$) ensures that the interpolation is balanced regardless of the scale of the raw feature space.

3.3 Experiment 2: Hybrid Kernel Results

3.3.1 Breast Cancer Wisconsin

Table 3 presents the alpha sweep results for the Breast Cancer dataset. The pure non-linear end ($\alpha = 0$) performs surprisingly poorly at 84.2%, while introducing even a small linear component ($\alpha = 0.1$) jumps performance to 97.4%. Peak accuracy of 98.3% is reached at $\alpha \in \{0.3, 0.5, 0.7, 0.8\}$, demonstrating that a moderate blend of linear and non-linear components is optimal for this dataset.

Table 3: Hybrid kernel alpha sweep — Breast Cancer Wisconsin dataset (114 test samples).

α	**Acc.**	**F1**	**Precision**	**Recall**	**AUC-ROC**
0.0	0.895	0.910	0.984	0.847	0.912
0.1	0.974	0.979	0.986	0.972	0.974
0.2	0.974	0.979	0.986	0.972	0.974
0.3	**0.982**	**0.986**	0.986	0.986	0.981
0.4	0.974	0.979	0.986	0.972	0.974
0.5	**0.982**	**0.986**	0.986	0.986	0.981
0.6	0.974	0.979	0.986	0.972	0.974
0.7	**0.982**	0.979	0.986	0.972	0.974
0.8	**0.982**	**0.986**	0.986	0.986	0.981
0.9	0.974	0.979	0.986	0.972	0.974
1.0	0.974	0.979	0.986	0.986	0.977

3.3.2 Chronic Kidney Disease

Table 4 shows results on the CKD dataset. At low alpha values ($\alpha \leq 0.2$), the model achieves a perfect test accuracy and F1 of 1.000, indicating the embedding captures a fully linearly separable representation in the non-linear subspace. Performance degrades gradually as α increases towards 1.0, settling at 96.3% accuracy and F1 of 0.949 for the pure linear kernel.

Table 4: Hybrid kernel alpha sweep — Chronic Kidney Disease dataset (80 test samples).

α	**Acc.**	**F1**	**Precision**	**Recall**	**AUC-ROC**
0.0	**1.000**	**1.000**	1.000	1.000	1.000
0.1	**1.000**	**1.000**	1.000	1.000	1.000
0.2	**1.000**	**1.000**	1.000	1.000	1.000
0.3	0.988	0.984	0.968	1.000	0.990
0.4	0.988	0.984	0.968	1.000	0.990
0.5	0.975	0.967	0.967	0.967	0.973
0.6	0.975	0.967	0.967	0.967	0.973
0.7	0.963	0.949	0.966	0.933	0.957
0.8	0.963	0.949	0.966	0.933	0.957
0.9	0.963	0.949	0.966	0.933	0.957
1.0	0.963	0.949	0.966	0.933	0.957

3.3.3 Adult Income

The Adult Income dataset presents a more challenging classification problem. Table 5 shows that the accuracy remains fairly flat across all alpha values (0.847–0.851). The F1 score peaks at $\alpha = 0.0$ (0.664) and declines gently to 0.650 at $\alpha = 1.0$. A 5-fold cross-validated linear SVM achieves a mean CV F1 of 0.656, confirming that the non-linear component adds marginal benefit over a well-tuned linear model on this dataset.

Table 5: Hybrid kernel alpha sweep — Adult Income dataset (test set).

α	**Accuracy**	**F1**
0.0	**0.851**	**0.664**
0.1	0.850	0.662
0.2	0.850	0.662
0.3	0.851	0.663
0.4	0.850	0.663
0.5	0.850	0.662
0.6	0.849	0.659
0.7	0.848	0.658
0.8	0.849	0.658
0.9	0.849	0.657
1.0	0.847	0.650

3.3.4 CDC Diabetes Health Indicators

The Diabetes dataset is the most challenging due to severe class imbalance (positive rate $\approx 11.2\%$). Table 6 reports balanced accuracy and macro F1. The best balanced accuracy (0.606) is achieved at $\alpha = 0.7$. The 5-fold CV linear SVM obtains a mean F1 of only 0.013, suggesting that even the non-linear component provides limited discriminative power, likely due to the low signal-to-noise ratio for the minority class.

Table 6: Hybrid kernel alpha sweep — CDC Diabetes dataset (10 000 test samples). Balanced accuracy and F1 reported.

α	**Balanced Accuracy**	**F1**
0.0	0.600	0.257
0.1	0.601	0.257
0.2	0.602	0.258
0.3	0.602	0.258
0.4	0.602	0.258
0.5	0.602	0.259
0.6	0.604	0.261
0.7	**0.606**	**0.262**
0.8	0.606	0.262
0.9	0.605	0.262
1.0	0.595	0.256

4 Discussion

4.1 Effectiveness of Explicit Embedding Pipelines

The results of Experiment 1 demonstrate that embedding transformations can meaningfully improve the performance of linear SVMs without replacing the kernel entirely. On the Ionosphere dataset, Kernel PCA produced the most discriminative representation, enabling a linear SVM to *exceed* the RBF SVM benchmark. This is a notable finding: by explicitly constructing a linearly separable manifold via Kernel PCA, the model inherits the interpretability benefits of a linear classifier while leveraging the representational power of a non-linear transformation.

Polynomial feature expansion, though it dramatically increases dimensionality (from 34 to 629 features), also matched the RBF SVM on accuracy, validating the theoretical equivalence between polynomial feature maps and polynomial kernels. The neural autoencoder and standard PCA provided moderate improvements, suggesting that reconstruction-based

objectives and linear variance-based projections are less aligned with class discriminability than kernel-guided or polynomial transformations.

4.2 The Performance–Interpretability Trade-off in Hybrid Kernels

Experiment 2 reveals a consistent trade-off between predictive performance and model interpretability as α varies:

- On the **Breast Cancer** dataset, the pure non-linear end ($\alpha = 0$) underperforms markedly (84.2%), while a modest linear component ($\alpha \geq 0.1$) recovers strong performance. This suggests that the embedding alone does not provide a well-structured latent space for this dataset, but combining it with the raw linear signal corrects this.
- On **CKD**, the embedding network produces a near-perfectly separable representation ($\alpha \leq 0.2$ yields 100% accuracy), confirming that the non-linear transformation finds the true class boundary. Increasing α predictably degrades performance as the model relies more on the raw linear signal.
- On the **Adult** dataset, performance is essentially flat across α, indicating that the data is intrinsically close to linearly separable for the features used. Embedding adds negligible benefit.
- On **Diabetes**, all configurations perform modestly due to class imbalance and the inherently noisy nature of self-reported health indicators. The flat performance curve suggests that neither linear nor non-linear representations can extract sufficient discriminative signal from this feature set alone.

4.3 Interpretability of Each Model

Interpretability answers three questions a practitioner cares about: *Why did the model predict this? Which features influenced the decision? How does changing a feature affect the output?* A model is interpretable when a human can answer these questions by inspecting the model directly, without needing extra tools.

Table 7 summarises whether each method studied in this project is interpretable and why.

Table 7: Interpretability of each model/embedding method.

Model	**Interpretable?**	**Reason**
Linear SVM	**Yes**	The weight vector $\mathbf{w}$ maps directly to original features. You can read off which feature pushes the prediction towards each class and by how much.
PCA + Linear SVM	**Partially**	$\mathbf{w}$ is readable in the PC space, and because PCA components are linear combinations of the original features, importance can be traced back to original features with one extra step.
Polynomial Features + Linear SVM	**Partially**	Weights exist for each term (e.g. x_1x_3), so interactions are explicit. However, with hundreds of terms it quickly becomes impractical to inspect manually.
Kernel PCA + Linear SVM	**No**	The kernel-induced components have no closed-form meaning in the original feature space. There is no way to trace a weight back to a specific input feature.
Autoencoder + Linear SVM	**No**	The latent dimensions are learned by the network and carry no human-readable meaning. You cannot answer "which input feature caused this?"
RBF SVM	**No**	There is no weight vector. The decision depends on distances to support vectors, making it impossible to link predictions to individual features.

4.4 Choosing a Model: Performance vs. Interpretability

For non-linearly structured data, the choice of embedding method should be guided by how much interpretability the application requires:

- **Interpretability is required** (e.g. healthcare, finance, legal) — use **PCA + Linear SVM**. It gains meaningful accuracy over a plain linear SVM and still allows a practitioner to trace predictions back to original features.

- **Interpretability is a secondary concern** — use **Kernel PCA + Linear SVM**. It achieved the best F1 in Experiment 1 (0.978), outperforming even the RBF SVM, at the cost of an opaque embedding space.
- **A tunable trade-off is needed** — use the **hybrid kernel** with parameter α. Setting α close to 1 keeps the model near a linear SVM (interpretable); reducing α gradually hands control to the non-linear embedding (better performance). This gives practitioners a single dial to navigate the trade-off continuously rather than committing to one extreme.

4.5 Practical Implications

Taken together, the two experiments suggest the following practical guidelines:

- If a dataset is moderately non-linear (e.g. Ionosphere, Breast Cancer), Kernel PCA or polynomial feature expansion followed by a linear SVM is a strong, interpretable alternative to an RBF SVM.
- The hybrid kernel parameter α provides a transparent lever for adjusting the interpretability–performance balance, particularly useful in high-stakes applications (e.g. healthcare) where explainability is required.
- When the data is heavily imbalanced or the feature–label relationship is weak (e.g. Diabetes), embedding-based enhancements provide limited benefit, and more fundamental data-level interventions (oversampling, better feature engineering) may be necessary first.

4.6 Limitations

Several limitations should be noted. The autoencoder in Experiment 1 was trained unsupervised, meaning its bottleneck embedding is not explicitly optimised for class discriminability. A supervised variant (e.g. a metric-learning approach) could yield stronger results. In Experiment 2, the hybrid kernel matrix is computed without hyperparameter tuning of the embedding network architecture, and γ for the RBF component is fixed. A joint optimisation of the embedding and kernel parameters could further improve performance. Finally, the CKD perfect-accuracy result ($\alpha \leq 0.2$) should be interpreted with caution given the small test set (80 samples).

5 Conclusion

This project investigated two strategies for extending linear kernel SVMs to non-linearly structured data through learned feature embeddings. The first experiment showed that

explicit embedding pipelines — particularly Kernel PCA and polynomial feature expansion — can close or even eliminate the performance gap between a linear and an RBF SVM on the UCI Ionosphere dataset. Kernel PCA achieved the highest F1 score of 0.978, surpassing the RBF benchmark by a small margin.

The second experiment introduced a hybrid kernel formulation (Equation 7) that linearly interpolates between a raw linear kernel and an embedding-induced non-linear kernel via a scalar α. Across four diverse structured datasets, the hybrid framework consistently revealed a smooth performance–interpretability trade-off. On the Breast Cancer and CKD datasets, the hybrid kernel achieved accuracies of up to 98.3% and 100% respectively. On the Adult Income and Diabetes datasets, the linear component was already near-optimal, reinforcing the value of linear models for tabular data with predominantly linear structure.

Collectively, these results support the thesis that learning adaptive representations for linear kernels is a viable and principled approach to structured data classification, offering practitioners a flexible framework that preserves interpretability while substantially improving accuracy over vanilla linear SVMs.

Future work will explore supervised embedding objectives, joint end-to-end training of the embedding and SVM, and extension to multi-class and regression settings.

Code and Results Availability

The implementation, notebooks, and source code for this project are available on GitHub:

GitHub Repository

The generated results, figures, and saved model outputs are available on Google Drive:

Google Drive Results Folder

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